POLITECNICO MILANO 1863 | DIPARTIMENTO DI ELETTRONICA INFORMAZIONE E BIOINGEGNERIA

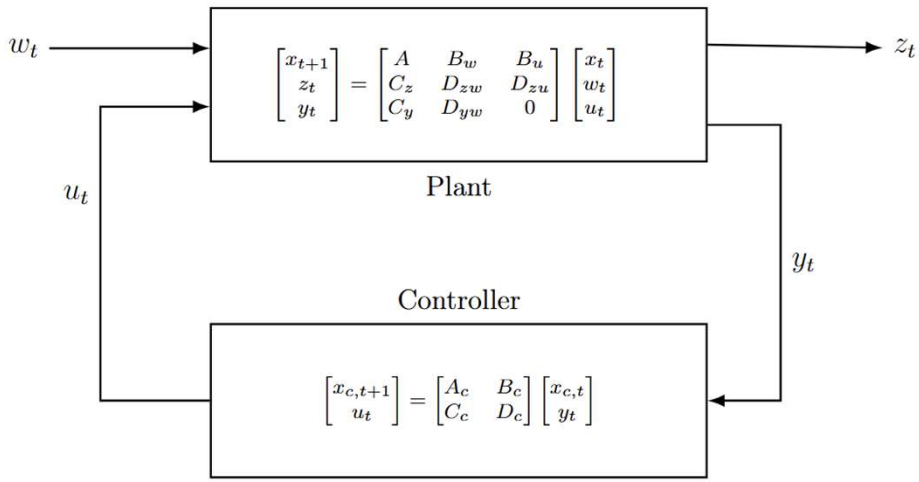
Direct Data-Driven

Distributionally Robust LMI synthesis for LTI systems

22.10.2025 I Giuseppe Fiengo

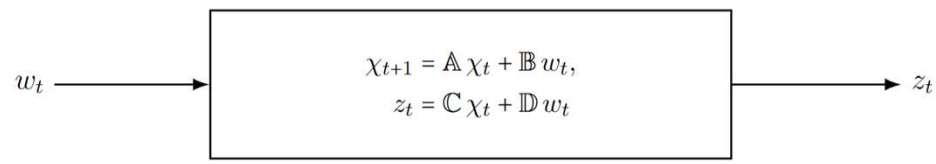
L2C
Learning to control

System (*linearized wind turbine blade control system*)



Define the combined state $\chi_t = [x_t^\top, x_{c,t}^\top]^\top$ and the closed-loop matrices

$$\mathbb{A} = \begin{bmatrix} A + B_u D_c C_y & B_u C_c \\ B_c C_y & A_c \end{bmatrix}, \quad \mathbb{B} = \begin{bmatrix} B_w + B_u D_c D_{yw} \\ B_c D_{yw} \end{bmatrix},$$
$$\mathbb{C} = \begin{bmatrix} C_z + D_{zu} D_c C_y & D_{zu} C_c \end{bmatrix}, \quad \mathbb{D} = D_{zw} + D_{zu} D_c D_{yw}.$$



Optimisation Problem

Baseline method

$$\min_{\theta \leftrightarrow (A_c, B_c, C_c, D_c)} J(\theta), \qquad \text{s.t. } \mathbb{A} \quad \text{isstable}$$

$$J(\theta) \;=\; \sup_{w \in \mathcal{W}} \quad \limsup_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\, \|z_t\|^2$$

Algorithm 1 Baseline Optimization via Monte-Carlo Cost

Require: Plant $(A, B_w, B_u, C_z, D_{zw}, D_{zu}, C_y, D_{yw})$, initial controller $(A_c^0, B_c^0, C_c^0, D_c^0)$, covariance $\Sigma_{\text{nom}} \succ 0$, seeds $\mathcal{S}$, horizon T, burn-in T_b

1: Parameterize $\theta \leftrightarrow (A_c, B_c, C_c, D_c)$ with simple box bounds.

2: Define stability test $\text{stable}(\mathbb{A}) \Leftrightarrow \rho(\mathbb{A}) < 1$.

3: Define penalized objective

$$\tilde{J}(\theta) = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \left[\frac{1}{T - T_b} \sum_{t=T_b}^{T-1} \|z_t^{(s)}\|^2 \right] \;+\; \lambda_\rho \max\{0, \rho(\mathbb{A}(\theta)) - 1\}^2.$$

4: For each seed $s \in \mathcal{S}$: simulate

$$\begin{aligned} y_t &= C_y x_t + D_{yw} w_t, \qquad & u_t &= C_c x_{c,t} + D_c y_t, \\ z_t &= C_z x_t + D_{zu} u_t + D_{zw} w_t, & & \\ x_{t+1} &= A x_t + B_u u_t + B_w w_t, \qquad & x_{c,t+1} &= A_c x_{c,t} + B_c y_t, \end{aligned}$$

with $w_t \sim \mathcal{N}(0, \Sigma_{\text{nom}})$.

5: Use L-BFGS-B (or similar) to compute $\theta^\star = \arg\min_\theta \tilde{J}(\theta)$.

6: If $\rho(\mathbb{A}(\theta^\star)) \geq 1$, apply a stabilizing projection to (A_c, D_c) and re-evaluate.

7: **return** optimized $(A_c^\star, B_c^\star, C_c^\star, D_c^\star)$ and the estimated cost.

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LMI method

$$\min_{\lambda\geq 0,\ X=X^\top,\ Y=Y^\top,\ K,L,M,N} \quad \mathrm{trace}(Q\Sigma_{\mathrm{nom}}) + \lambda\,\gamma^2$$

$$\begin{pmatrix} T^\top \mathcal{P} A T & T^\top \mathcal{P} B \\ CT & D \end{pmatrix} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}, \qquad T^\top \mathcal{P} T = P.$$

$$\mathbb{P} = \begin{bmatrix} Y & I \\ I & X \end{bmatrix}, \qquad \mathbf{A} = \begin{bmatrix} AY + B_u M & A + B_u N C_y \\ K & XA + LC_y \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} B_w + B_u N D_{yw} \\ XB_w + LD_{yw} \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} C_z Y + D_{zu} M & C_z + D_{zu} N C_y \end{bmatrix}, \quad \mathbf{D} = D_{zw} + D_{zu} N D_{yw}.$$

Correlated model ($\mathcal{W}_{\mathrm{cor}}$). With identity blocks I_w, I_z of matching sizes, enforce

$$\begin{bmatrix} -\mathbb{P} & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ 0 & \lambda I_w & -Q - \lambda I_w & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -\mathbb{P} & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0, \qquad Q \succeq 0, \quad \mathbb{P} \succeq 0.$$

Independent model ($\mathcal{W}_{\mathrm{ind}}$). Impose the pair of LMIs

$$\begin{bmatrix} -\mathbb{P} & \mathbf{A}^\top & \mathbf{C}^\top \\ \mathbf{A} & -\mathbb{P} & 0 \\ \mathbf{C} & 0 & -I_z \end{bmatrix} \prec 0, \qquad \begin{bmatrix} -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ \lambda I_w & -Q - \lambda I_w & 0 & 0 \\ \mathbf{B} & 0 & -\mathbb{P} & 0 \\ \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0, \quad Q \succeq 0,\ \mathbb{P} \succeq 0.$$

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Direct Data-Driven (DDD) approach

State matrices

$$X_- = \begin{bmatrix} x_0 & \cdots & x_{T-1} \end{bmatrix}, \qquad U_- = \begin{bmatrix} u_0 & \cdots & u_{T-1} \end{bmatrix}, \qquad X_+ = \begin{bmatrix} x_1 & \cdots & x_T \end{bmatrix}.$$

$$D_x = \begin{bmatrix} X_- \\ U_- \end{bmatrix} \qquad \underbrace{\begin{bmatrix} \hat{A} & \hat{B}_u \end{bmatrix}}_{\hat{O}_x} = X_+ D_x^\dagger,$$

Disturbance matrix: $R := X_+ - \hat{A}X_- - \hat{B}_u U_- \in \mathbb{R}^{n \times T}.$

$$S_R := \frac{1}{T} R R^\top, \quad \tilde{S}_R := \tfrac{1}{2}\left(S_R + S_R^\top\right) + \varepsilon I_n = U_s \operatorname{diag}(s_1, \ldots, s_n) U_s^\top,$$

$$n_w := \min\left\{k : \textstyle\sum_{i=1}^k s_i \geq \eta \sum_{i=1}^n s_i\right\} \quad \widehat{\Sigma}_w := \operatorname{diag}(s_1, \ldots, s_{n_w})$$

$$\hat{B}_w \widehat{\Sigma}_w^{1/2} := U_s(:, 1{:}n_w) \operatorname{diag}\left(\sqrt{s_1}, \ldots, \sqrt{s_{n_w}}\right), \qquad \hat{w} := \hat{B}_w^\dagger R,$$

$$\Sigma_{\text{nom}} := \tfrac{1}{2}\left(\tfrac{1}{T}\hat{w}\hat{w}^\top + \left(\tfrac{1}{T}\hat{w}\hat{w}^\top\right)^\top\right) + \varepsilon I_{n_w}.$$

Analogously, with $D_y = \begin{bmatrix} X_- \\ \hat{w} \end{bmatrix}$ and $D_z = \begin{bmatrix} X_- \\ U_- \\ \hat{w} \end{bmatrix}$,

$$\underbrace{\begin{bmatrix} \hat{C}_y & \hat{D}_{yw} \end{bmatrix}}_{\hat{O}_y} = Y_- D_y^\dagger, \qquad \underbrace{\begin{bmatrix} \hat{C}_z & \hat{D}_{zu} & \hat{D}_{zw} \end{bmatrix}}_{\hat{O}_z} = Z_- D_z^\dagger.$$

Matrix Errors

Consider the matrix errors: $\|\Delta_x\|_F \leq \beta \quad \text{with} \quad \beta := \dfrac{\|R\|_F}{\max(\sigma_{\min}, \epsilon)}.$

$$\begin{bmatrix} A & B_u \end{bmatrix} = \underbrace{\begin{bmatrix} \hat{A} & \hat{B}_u \end{bmatrix}}_{\hat{O}_x} + \Delta_x, \quad \Delta_x = \begin{bmatrix} \Delta_A & \Delta_B \end{bmatrix}, \quad \|\Delta_i\|_F \leq \beta_i$$

$$\beta_A = \beta\sqrt{\frac{n_x}{n_x + n_u}}, \quad \beta_B = \beta\sqrt{\frac{n_u}{n_x + n_u}}$$

Thus: $\mathbf{A}_{\text{true}} = \mathbf{A}(\hat{A}, \hat{B}_u) + E_{AA} + E_{AB},$

$$E_{AA} = \begin{bmatrix} \Delta_A Y & \Delta_A \\ 0 & X\Delta_A \end{bmatrix} \qquad E_{AB} = \begin{bmatrix} \Delta_B M & \Delta_B N C_y \\ 0 & 0 \end{bmatrix}$$

$$\boxed{\|E_{AA}\|_F \leq \sqrt{1 + \|X\|_2^2 + \|Y\|_2^2}\,\beta_A = \beta_{AA}, \qquad \|E_{AB}\|_F \leq \sqrt{\|M\|_2^2 + \|NC_y\|_2^2}\,\beta_B = \beta_{AB}.}$$

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Augmented LMI problem

$$\min_{\substack{\lambda\geq 0,\ Q,\\ \{s_{AA,i}\geq 0,\ \tau_{AA,i}\geq 0\}_{i\in(1,n_x)},\\ s_{AB}\geq 0,\ \tau_{AB}\geq 0,\ \{t\}_{i\in(K,L,M,N)}\\ X=X^T,\ Y=Y^T,\ K,\ L,\ M,\ N}} \quad \mathrm{tr}(Q\,\Sigma_{\mathrm{nom}}) + \lambda\gamma^2 + \sum_{j\in(A,B)} \mu_{A,j} \sum_{i=1}^{n_x}\Big(s_{A,j,i} + \frac{\tau_{A,j,i}}{\beta_{A,j,i}^2+\varepsilon}\Big) + \sum_{i\in(K,K,M,N)} \rho_i\,(t_i+\varepsilon)$$

$$\text{s.t. } P = \begin{bmatrix} Y & I \\ I & X \end{bmatrix} \succ 0, \quad \Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D})\Big|_{(4,4)} \leftarrow -P\Big(\sum_{i=1}^{n_x}\tau_{AA,i} + \tau_{AB}\Big) I_{2n_x},$$

$$\begin{bmatrix} \Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}) & \Pi_{AA} & \Pi_{AB} \\ \Pi_{AA}^\top & -\mathrm{Diag}(s_{AA}) & 0 \\ \Pi_{AB}^\top & 0 & -s_{AB} I_{n_x} \end{bmatrix} \prec 0,$$

$$\begin{bmatrix} s_{AA,i} & \beta_{AA,i} \\ \beta_{AA,i} & \tau_{AA,i} \end{bmatrix}\Bigg|_{i\in(1,n_x)} \succeq 0, \qquad \begin{bmatrix} s_{AB} & \beta_{AB} \\ \beta_{AB} & \tau_{AB} \end{bmatrix} \succeq 0,$$

$$\begin{bmatrix} t_K I_{n_x} & 0 & 0 & 0 & K \\ 0 & t_L I_{n_x} & 0 & 0 & L \\ 0 & 0 & t_M I_{n_u} & 0 & M \\ 0 & 0 & 0 & t_N I_{n_u} & N \\ K^\top & L^\top & M^\top & N^\top & \mathrm{Diag}(t_K I_{n_x}, t_L I_{n_y}, t_M I_{n_x}, t_N I_{n_y}) \end{bmatrix} \succeq 0.$$

$$\Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}) := \begin{bmatrix} -P & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & -\lambda I & \lambda I & \mathbf{B}^\top & \mathbf{D}^\top \\ 0 & \lambda I & -Q-\lambda I & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -P & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I \end{bmatrix} \prec 0,$$

$$2\langle E_{A,i}, Z\rangle \;\leq\; s_{A,i}\,\beta_{A,i}^2 + \frac{1}{s_{A,i}}\|Z\|_F^2, \qquad Z = P\mathbf{A},$$

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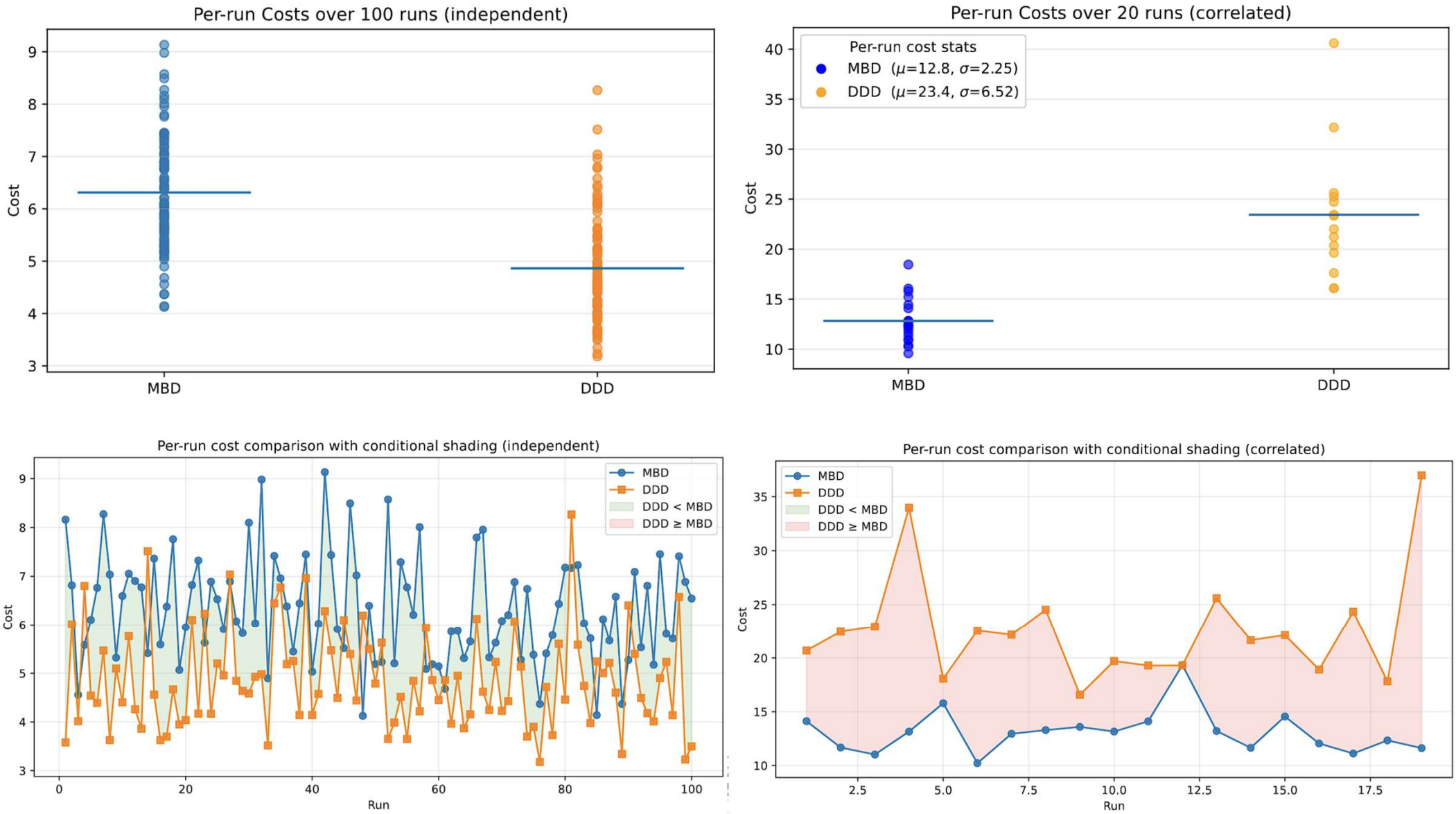
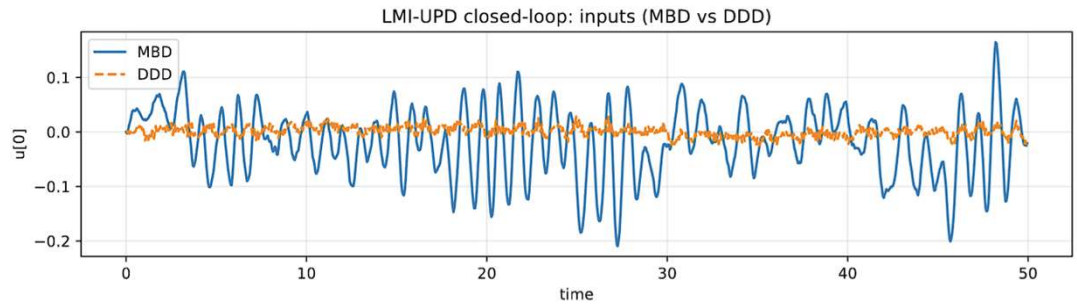
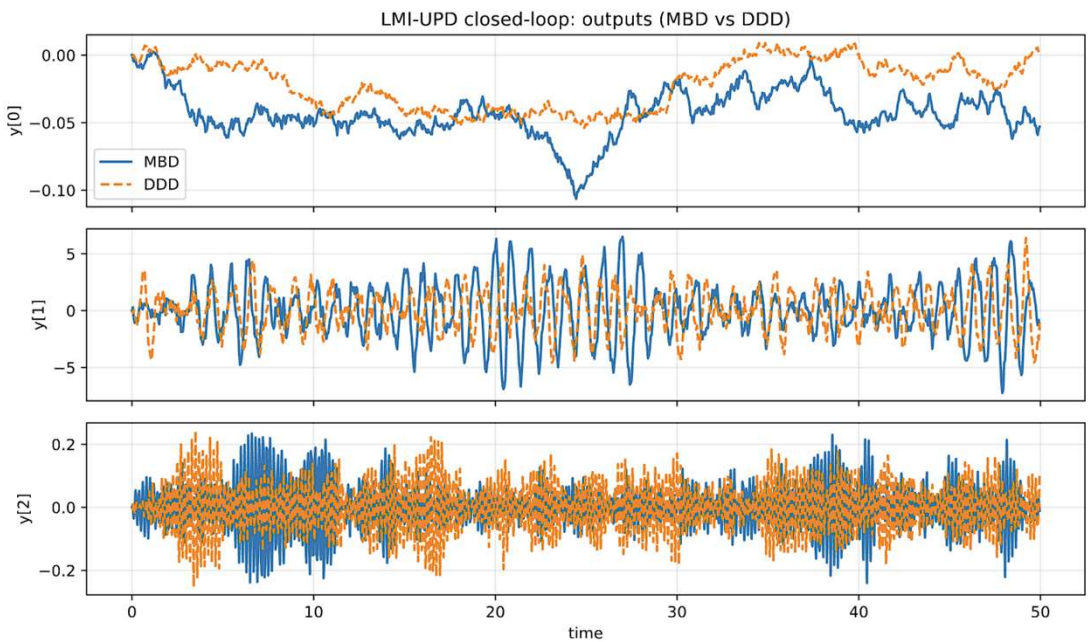
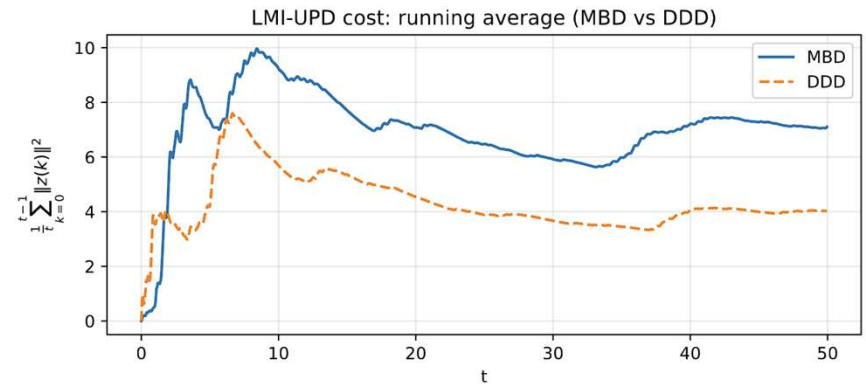
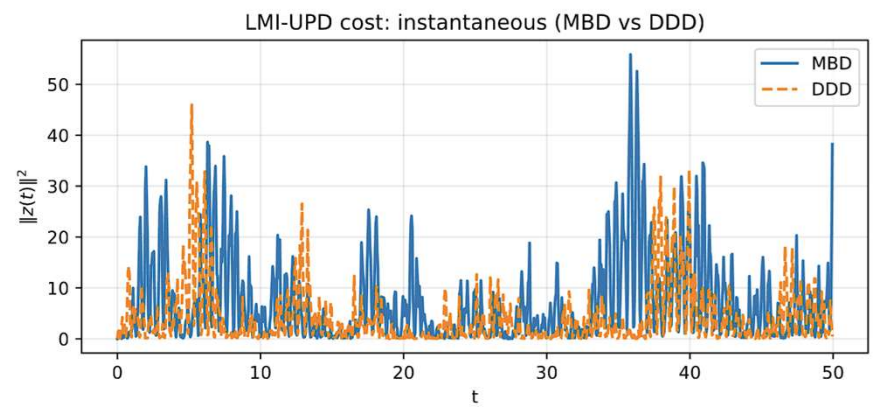
Simulation Results (MBD vs DDD)

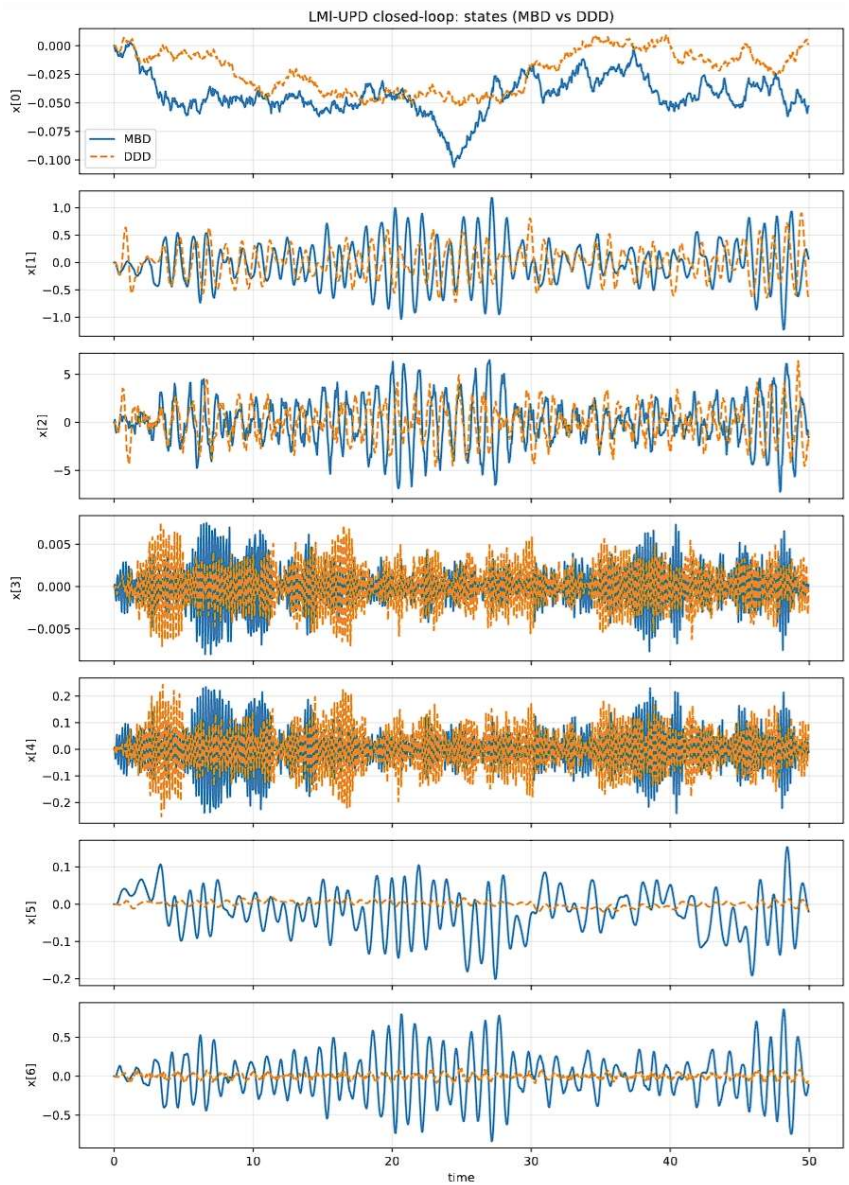

Per-run Costs over 100 runs (independent)
Cost
MBD
DDD
Per-run Costs over 20 runs (correlated)
Per-run cost stats
MBD (μ=12.8, σ=2.25)
DDD (μ=23.4, σ=6.52)
Cost
MBD
DDD
Per-run cost comparison with conditional shading (independent)
MBD
DDD
DDD < MBD
DDD ≥ MBD
Cost
Run
Per-run cost comparison with conditional shading (correlated)
MBD
DDD
DDD < MBD
DDD ≥ MBD
Cost
Run

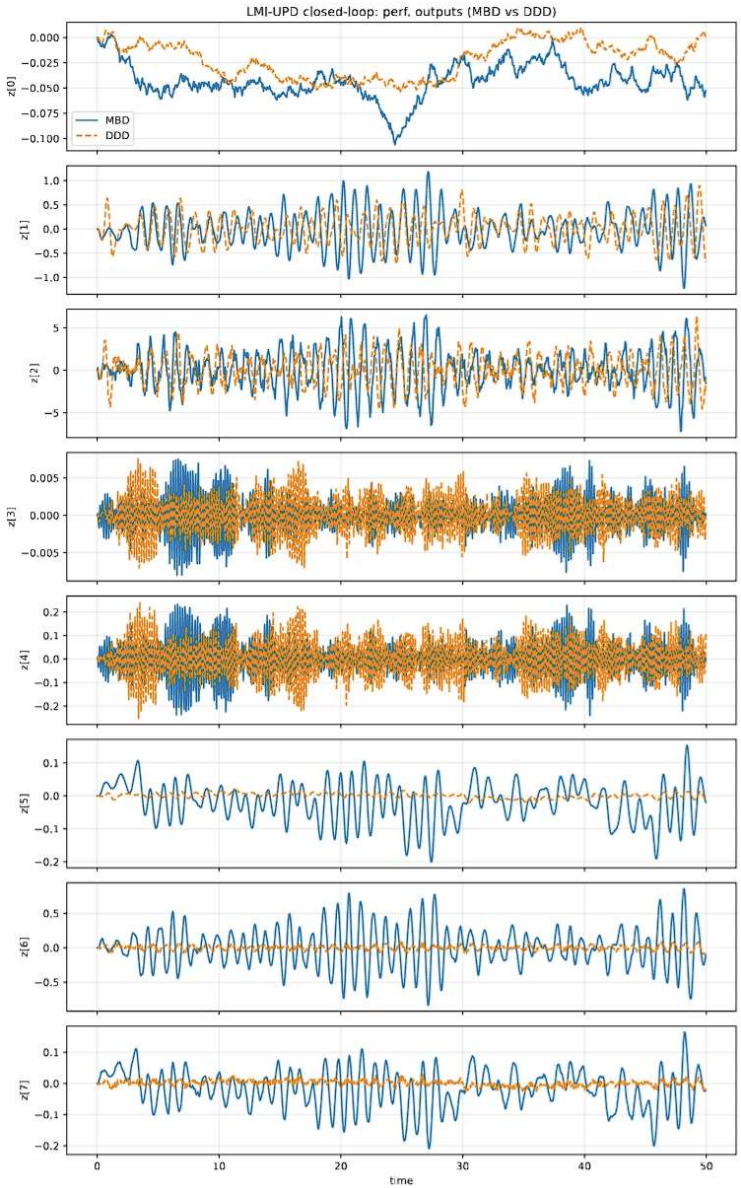

LMI-UPD closed-loop: inputs (MBD vs DDD)
MBD
DDD
u[0]
time


LMI-UPD closed-loop: outputs (MBD vs DDD)
MBD
DDD
y[0]
y[1]
y[2]
time


LMI-UPD cost: running average (MBD vs DDD)
MBD
DDD
t


LMI-UPD cost: instantaneous (MBD vs DDD)
MBD
DDD
t


LMI-UPD closed-loop: states (MBD vs DDD)
MBD
DDD
x[0]
x[1]
x[2]
x[3]
x[4]
x[5]
x[6]
time


LMI-UPD closed-loop: perf. outputs (MBD vs DDD)
MBD
DDD
z[0]
z[1]
z[2]
z[3]
z[4]
z[5]
z[6]
z[7]
time

Unresolved

Do a sensitivity analysis to adjust the constraints when simulating correlated disturbances.

Understand why MOSEK fails with the new problem formulation.

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Contatti

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Learning to control